Report: Machine Learning-Based Sentiment Analysis

1. Objective

In this part, we transition from rule-based counting to **Supervised Machine Learning**. We trained models to recognize sentiment patterns in three classes: **Positive (+1), Neutral (0), and Negative (-1)**.

2. Feature Engineering & Setup

- **Train-Test Split:** We used a **70-30 split** as required. 13,031 samples were used for training and 5,586 for testing.
- **Vectorization (TF-IDF):** We used `TfidfVectorizer` with `ngram_range=(1, 2)`.
    - **Justification:** TF-IDF weights words based on their importance. Including **bigrams** (2-word sequences) allows the model to understand context, such as "stock crash" or "interest hike," which carry more meaning than individual words.

3. Model Outcomes

We compared **Multinomial Naive Bayes (NB)** and **Logistic Regression (LR)**. Below are the results from the `classification_report` output:

Model A: Naive Bayes

- **Overall Accuracy:** 61.49%
- **Observation:** Performed very well on **Neutral** classes (0.83 F1-score) but struggled to distinguish between the directional sentiments of Positive and Negative.

Model B: Logistic Regression

- **Overall Accuracy: 62.28%**
- **Observation:** This was the **best-performing model**. It showed superior performance in the **Neutral** class with a **0.86 F1-score**.

Detailed Comparison Table

Class	NB F1-Score	LR F1-Score
Negative (-1)	0.48	0.47
Neutral (0)	0.83	**0.86**

Positive (+1)	0.54	0.54
Overall Accuracy	61.49%	**62.28%**

4. Conclusion

The results clearly indicate that **Logistic Regression** is more robust for financial sentiment analysis. While both models struggled with directional labels (Positive/Negative) due to the complexity of financial jargon, Logistic Regression's ability to handle overlapping features led to a higher overall performance.
